Selection Games with Compact Sets

C. Caruvana
with J. Holshouser

Indiana University Kokomo

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Introduction

As a general theme, we find connections between selection principles on a space X and related selection principles on hyperspaces of X. We elaborate on these connections via selection games, which generalize selection principles.

We will be working with single- and finite-selection games involving classes $\mathcal{A}$ and $\mathcal{B}$, which are denoted $\mathsf{G}_1(\mathcal{A},\mathcal{B})$ and $\mathsf{G}_{\mathrm{fin}}(\mathcal{A},\mathcal{B})$, respectively. The finite-selection game is played as follows.

- In the nth inning, One selects $A_n \in \mathcal{A}$ and
- Two responds with $\mathcal{F}_n \in [A_n]^{<\omega}$.

We declare Two the winner if $\bigcup\{\mathcal{F}_n : n \in \omega\} \in \mathcal{B}$.

The game $\mathsf{G}_1(\mathcal{A},\mathcal{B})$ is defined in the analogous way.

The various kinds of strategies for single-selection games are as follows.

- A *strategy for player One* in $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ is a function $\sigma : (\bigcup \mathcal{A})^{<\omega} \to \mathcal{A}$. A strategy σ for One is called *winning* if whenever $x_n \in \sigma\langle x_k : k < n\rangle$ for all $n \in \omega$, $\{x_n : n \in \omega\} \notin \mathcal{B}$. If player One has a winning strategy, we write $\mathrm{I} \uparrow \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.
- A *strategy for player Two* in $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ is a function $\tau : \mathcal{A}^{<\omega} \to \bigcup \mathcal{A}$. A strategy τ for Two is *winning* if whenever $A_n \in \mathcal{A}$ for all $n \in \omega$, $\{\tau(A_0, \ldots, A_n) : n \in \omega\} \in \mathcal{B}$. If player Two has a winning strategy, we write $\mathrm{II} \uparrow \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.

- A *predetermined strategy* for One is a strategy which only considers the current turn number. Formally it is a function $\sigma : \omega \to \mathcal{A}$. If One has a winning predetermined strategy, we write $\mathrm{I} \underset{\text{pre}}{\uparrow} \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.
- A *Markov strategy* for Two is a strategy which only considers the most recent move of player One and the current turn number. Formally it is a function $\tau : \mathcal{A} \times \omega \to \bigcup \mathcal{A}$. If Two has a winning Markov strategy, we write $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.

- If there is a single element $x_0 \in \mathcal{A}$ so that the constant function with value x_0 is a winning strategy for One, we say that One has a *constant winning strategy*, denoted by $\mathrm{I} \underset{\text{cnst}}{\uparrow} \mathsf{G}_1(\mathcal{A}, \mathcal{B})$.

All of these have obvious analogs for the finite-selection games.

So how do these strategy types relate to selection principles?

Let's start with Lindelöf-like selection principles, referred to as Bar-Ilan selection principles in Scheepers 2003.

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For collections $\mathcal{A}$ and $\mathcal{B}$, $\binom{\mathcal{A}}{\mathcal{B}}$ means that, for every $A \in \mathcal{A}$, there exists $B \subseteq A$ so that $B \in \mathcal{B}$.

As pointed out in Clontz 2020,

Lemma

For any space X,

$$X \models \begin{pmatrix} \mathcal{A} \\ \mathrm{ctbl}(\mathcal{B}) \end{pmatrix} \iff \mathrm{I} \underset{\mathrm{cnst}}{\not\uparrow} \mathsf{G}_1(\mathcal{A}, \mathcal{B}) \iff \mathrm{I} \underset{\mathrm{cnst}}{\not\uparrow} \mathsf{G}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B})$$

where $\mathrm{ctbl}(\mathcal{B}) = \{B \in \mathcal{B} : |B| \leqslant \omega\}$.

That is, constant strategies for One characterize Bar-Ilan selection principles.

The standard selection principles are the single- and finite-selection principles corresponding to collections $\mathcal{A}$ and $\mathcal{B}$, denoted by $\mathsf{S}_1(\mathcal{A}, \mathcal{B})$ and $\mathsf{S}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B})$, respectively.

Recall that $X \models \mathsf{S}_1(\mathcal{A}, \mathcal{B})$ if, for any sequence $\langle A_n : n \in \omega \rangle \in \mathcal{A}^\omega$, there exists for each n some $x_n \in A_n$ so that $\{x_n : n \in \omega\} \in \mathcal{B}$. The finite-selection principle is similarly defined.

Clontz 2020 also notes that

Lemma

$$X \models \mathsf{S}_{\square}(\mathcal{A}, \mathcal{B}) \iff \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathsf{G}_{\square}(\mathcal{A}, \mathcal{B})$$

where $\square \in \{1, \mathrm{fin}\}$.

That is, predetermined strategies for One characterize selection principles.

For a given space X, we consider

- $\mathscr{T}_X$ to be the class of all proper, non-empty open subsets of X.
- non-trivial open covers: all open covers $\mathscr{U}$ so that $X \notin \mathscr{U}$. We let $\mathcal{O}_X$ denote all non-trivial open covers of X.
- ω-covers: those open covers $\mathscr{U}$ so that, for any $F \in [X]^{<\omega}$, there exists $U \in \mathscr{U}$ so that $F \subseteq U$. We let Ω_X denote all ω-covers of X.
- k-covers: those open covers $\mathscr{U}$ so that, for any compact $K \subseteq X$, there exists $U \in \mathscr{U}$ so that $K \subseteq U$. We let $\mathcal{K}_X$ denote all k-covers of X.

Some particular selections principles are

- $\begin{pmatrix} \mathcal{O} \\ \mathrm{ctbl}(\mathcal{O}) \end{pmatrix}$, the Lindelöf property.
- $\begin{pmatrix} \Omega \\ \mathrm{ctbl}(\Omega) \end{pmatrix}$, the ω-Lindelöf property, most commonly known as the ϵ-space property.
- $\begin{pmatrix} \mathcal{K} \\ \mathrm{ctbl}(\mathcal{K}) \end{pmatrix}$, the k-Lindelöf property.
- $\mathsf{S}_{\mathrm{fin}}(\mathcal{O}, \mathcal{O})$, the Menger property.
- $\mathsf{S}_1(\mathcal{O}, \mathcal{O})$, the Rothberger property.

Background and Inspiration

For a given space X, let $X^{<\omega}$ denote the disjoint union of X^n for all $n \geqslant 1$. Consider the fact that $X^{<\omega}$ is a topological coding space for finite subsets of X. Hence, we could expect some relationship between open covers of $X^{<\omega}$ and ω-covers of X.

An important tool here is the ability to refine ω-covers on finite products of X into cubes that correspond to ω-covers on X.

Lemma (Just, Miller, Scheepers 1996)

Let X be a space and $n \geqslant 1$. Then,

- if $\mathscr{U}$ is an ω-cover of X, then $\{U^n : U \in \mathscr{U}\}$ is an ω-cover of X^n.
- if $\mathscr{U}$ is an ω-cover of X^n,

$$\mathscr{V} = \{V \in \mathscr{T}_X : (\exists U \in \mathscr{U})[V^n \subseteq U]\}$$

  is an ω-cover of X.

The connections between ω-covers on X and open covers on $X^{<\omega}$ are stated below.

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Theorem (Gerlits, Nagy 1982)

Let X be a space. The following are equivalent:

- X is an ϵ-space,
- every finite power of X is an ϵ-space,
- every finite power of X is Lindelöf, and
- $X^{<\omega}$ is Lindelöf.

Theorem (Just, Miller, Scheepers 1996)

Let X be a space. The following are equivalent:

- $X \models \mathsf{S}_{\mathrm{fin}}(\Omega, \Omega)$,
- $(\forall n \in \omega)[X^{n+1} \models \mathsf{S}_{\mathrm{fin}}(\Omega, \Omega)]$,
- $(\forall n \in \omega)[X^{n+1} \models \mathsf{S}_{\mathrm{fin}}(\mathcal{O}, \mathcal{O})]$, and
- $X^{<\omega}$ is Menger.

Theorem (Sakai 1988)

Let X be a space. The following are equivalent:

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- $(\forall n \in \omega)[X^{n+1} \models \mathsf{S}_1(\Omega, \Omega)]$,
- $(\forall n \in \omega)[X^{n+1} \models \mathsf{S}_1(\mathcal{O}, \mathcal{O})]$, and
- $X^{<\omega}$ is Rothberger.

As it turns out, all three of the following theorems can be generalized to the corresponding selection games. In particular,

Theorem

The games

- $\mathsf{G}_{\square}(\Omega_X, \Omega_X)$,
- $\mathsf{G}_{\square}(\Omega_{X^{<\omega}}, \Omega_{X^{<\omega}})$, and
- $\mathsf{G}_{\square}(\mathcal{O}_{X^{<\omega}}, \mathcal{O}_{X^{<\omega}})$

where $\square \in \{1, \mathrm{fin}\}$ are equivalent with respect to all of the strategy levels mentioned above.

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where $\square \in \{1, \mathrm{fin}\}$ are equivalent with respect to all of the strategy levels mentioned above.

Can we find a similar correspondence for games involving k-covers?

Results

Since we will be investigating mostly finite sets and compact sets, we identify an adequate space to capture these sets.

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For a given topological space X, let $\mathbb{K}(X)$ denote the space of compact subsets endowed with the **Vietoris topology**, which has as its basis sets of the form

$$[U_1, \ldots, U_n] = \left\{ K \in \mathbb{K}(X) : (\forall j)(K \cap U_j \neq \emptyset) \wedge \left(K \subseteq \bigcup_{j=1}^{n} U_j \right) \right\}.$$

We also distinguish

$$\mathcal{P}_{\text{fin}}(X) = \{F \subseteq X : F \in [X]^{<\omega}\}$$

viewed as a subspace of $\mathbb{K}(X)$. In fact,

Theorem

The games

- $\mathsf{G}_{\square}(\Omega_X, \Omega_X)$,
- $\mathsf{G}_{\square}(\Omega_{\mathcal{P}_{\text{fin}}(X)}, \Omega_{\mathcal{P}_{\text{fin}}(X)})$, and
- $\mathsf{G}_{\square}(\mathcal{O}_{\mathcal{P}_{\text{fin}}(X)}, \mathcal{O}_{\mathcal{P}_{\text{fin}}(X)})$

where $\square \in \{1, \text{fin}\}$ are equivalent with respect to all of the strategy levels mentioned above.

The key difference in this context as compared to the finite powers context is the avoidance of the combinatorial issues that arise with finite powers. What's more, is the proof of the equivalence in this context is directly analogous to the case when we consider compact subsets.

We first establish that k-covers of $\mathbb{K}(X)$ can be refined into a convenient form of covers that correspond directly to k-covers of X.

Lemma

Let X be a space. Then,

- if $\mathscr{U}$ is a k-cover of X, then $\{[U] : U \in \mathscr{U}\}$ is a k-cover of $\mathbb{K}(X)$.
- if $\mathscr{U}$ is a k-cover of $\mathbb{K}(X)$, then

$$\mathscr{V} = \{V \in \mathscr{T}_X : \exists U \in \mathscr{U}([V] \subseteq U)\}$$

  is a k-cover of X.

Sketch of Proof.

Suppose $\mathscr{U}$ is a k-cover of X and let $\mathbf{K} \subseteq \mathbb{K}(X)$ be compact. As $\bigcup \mathbf{K}$ is a compact subset of X, there is some $U \in \mathscr{U}$ so that $\bigcup \mathbf{K} \subseteq U$. Hence, $\mathbf{K} \subseteq [U]$ and this claim is proved.

Sketch of Proof.

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Suppose $\mathscr{U}$ is a k-cover of $\mathbb{K}(X)$ and let $K \subseteq X$ be compact. Since $K^* := \{L \in \mathbb{K}(X) : L \subseteq K\}$ is a compact subset of $\mathbb{K}(X)$, we can find $U \in \mathscr{U}$ so that $K^* \subseteq U$. For each $L \in K^*$, let $\mathscr{B}_L$ be a basic neighborhood so that $L \in \mathscr{B}_L \subseteq U$. By compactness, we can find $L_1, \ldots, L_n$ so that

$$K^* \subseteq \bigcup_{j=1}^{n} \mathscr{B}_{L_j}.$$

Let $\mathscr{B}_{L_j} = [W_{j,1}, W_{j,2}, \ldots, W_{j,m_j}]$ for each j and, for each $x \in K$, let

$$N_x = \bigcap \{W_{j,k} : x \in W_{j,k}\}.$$

Then, $V := \bigcup_{x \in K} N_x$ is as desired. $\square$

Using the above correspondence between k-covers and the methods of strategy translation, we obtain

Theorem

The games

- $\mathsf{G}_{\mathrm{fin}}(\mathcal{K}_X, \mathcal{K}_X)$,
- $\mathsf{G}_{\mathrm{fin}}(\mathcal{K}_{\mathbb{K}(X)}, \mathcal{K}_{\mathbb{K}(X)})$, and
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are equivalent with respect to all of the strategy levels mentioned above.

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are equivalent with respect to all of the strategy levels mentioned above.

However, we cannot guarantee equivalence at the single-selection level.

Some notable corollaries are:

- The Hurewicz theorem holds:

$$\mathrm{I} \underset{\text{pre}}{\uparrow} \mathsf{G}_{\text{fin}}(\mathcal{K}_X, \mathcal{K}_X) \iff \mathrm{I} \uparrow \mathsf{G}_{\text{fin}}(\mathcal{K}_X, \mathcal{K}_X).$$

- X is k-Lindelöf iff $\mathbb{K}(X)$ is k-Lindelöf iff $\mathbb{K}(X)$ is Lindelöf.
- X is k-Menger iff $\mathbb{K}(X)$ is k-Menger iff $\mathbb{K}(X)$ is Menger.

Though we do not obtain the full equivalence at the single-selection level, we still obtain Pawlikowski's theorem.

Theorem

$$\mathrm{I} \underset{\text{pre}}{\uparrow} \mathsf{G}_1(\mathcal{K}_X, \mathcal{K}_X) \iff \mathrm{I} \uparrow \mathsf{G}_1(\mathcal{K}_X, \mathcal{K}_X).$$

Thank You!